

SafeRoute AI — Project Summary

AI-Powered Safe Navigation & Multimodal Risk Explainability Platform for Nighttime Mobility

Repository: github.com/harshal9834/roshani_hackathon

Tech Stack: React 19, Vite, Tailwind CSS v4, TypeScript

Focus: Women's Nighttime Transit Safety

Key AI Module: SHAP Feature Attribution & Anomaly Rerouting

1. Executive Summary & Problem Vision

SafeRoute AI is an intelligent navigation platform engineered to address the critical safety challenges women face when commuting after dark. Traditional navigation applications optimize strictly for distance and traffic velocity, often routing pedestrians through unlit alleyways, isolated pathways, or hazard zones. SafeRoute AI introduces **multidimensional safety scoring**, combining real-time municipal street lighting telemetry, pedestrian footfall density, transit proximity, and historical incident intelligence to compute verifiable safe corridors.

2. Core AI & Safety Innovations

1. Multi-Signal Safety Engine	Computes real-time 0-100% safety indices using weighted parameters: continuous Lux street lighting, crowd activity index, 24/7 CCTV surveillance, and proximity to active police kiosks.
2. Explainable AI (SHAP Waterfall)	Solves the AI 'black box' dilemma by visually breaking down feature contributions: +35 pts Street Lighting, +22 pts Crowd Activity, +15 pts Public Transit, penalized by -12 pts for nearby incident reports.
3. Dynamic Anomaly Re-routing	Continuously monitors telemetry during live navigation. If a lighting failure or hazard is detected ahead, the app instantly alerts the user and recalculates an optimal safe bypass route.
4. Emergency SOS & Refuge Hub	Features a 5-second countdown panic trigger, live GPS coordinate dispatch to trusted guardians, and a locator for verified 24/7 safe refuges (Police stations, 24/7 pharmacies, metro plazas).
5. AI Safety Copilot	An embedded natural language assistant that answers real-time safety inquiries (e.g., 'Is Grand Blvd safe right now?', 'Where is the nearest help booth?').

3. Complete 14-Screen Architecture & Feature Matrix

Screen / Module	Component File	Functionality & Key Highlights
1. Splash Screen	SplashScreen.tsx	Branded onboarding showcasing core safety promises, AI verification, and start CTA.
2. Login & Auth	LoginScreen.tsx	Phone OTP authentication with biometric toggle and trusted guardian setup.
3. Dashboard	DashboardScreen.tsx	Central command hub: Live 94% safety index, night mode toggle, and quick SOS button.
4. Route Search	RouteSearchScreen.tsx	Origin & destination search with filters for lit paths, metro lines, and guarded zones.
5. Route Comparison	RouteComparisonScreen.tsx	Compares Route A (58%), Route B (79%), and Route C (93% Recommended safe path).
6. SHAP Explainability	ShapExplainabilityScreen.tsx	Interactive XAI waterfall attribution chart displaying positive and negative feature weights.
7. Live Navigation	LiveNavigationScreen.tsx	Turn-by-turn HUD with step safety notes, continuous lux meter, and telemetry markers.
8. Safety Alert Modal	SafetyAlertModal.tsx	Instant pop-up alert for sudden hazard or dark zone entry with one-click recalculate.
9. Dynamic Re-routing	DynamicReroutingScreen.tsx	Side-by-side comparison of original vs. newly recalculated safe bypass corridor.
10. Safety Check-In	SafetyCheckInScreen.tsx	Automated periodic check-in prompt: 'I Am Safe' acknowledgment or instant SOS escalation.
11. Emergency SOS	EmergencySosScreen.tsx	5-sec countdown panic button, siren trigger, live GPS coordinate dispatch, and refuge list.
12. Trusted Contacts	TrustedContactsScreen.tsx	Guardian management with real-time location sharing status and phone battery indicators.
13. Safety Analytics	SafetyAnalyticsScreen.tsx	Weekly safety summaries, safety trend by hour (8 PM - 1 AM), and risk factor breakdown.
14. Profile & Settings	ProfileSettingsScreen.tsx	Night mode configuration, sensitivity sliders, offline safe maps, and guardian prefs.

4. Core Reusable Infrastructure & Simulation Engine

- **MapEngine.tsx:** Custom SVG vector map canvas supporting multi-layer risk heatmaps, polyline routing, user geolocation beacons, and interactive emergency shelter pins.
- **MobileFrame.tsx:** Hardware bezel emulator with an integrated developer sidebar for switching between all 14 screens and triggering simulated hazard events.
- **FloatingAiAssistant.tsx:** Embedded AI Copilot with instant prompt chips and telemetry parsing.
- **mockData.ts:** Seed data structures for routes, SHAP scores, emergency help points, and analytical metrics.

5. Execution & Build Commands

Command	Purpose
npm run dev	Starts the local Vite development server on port 3000 with hot module replacement.
npm run lint	Executes TypeScript type verification (tsc --noEmit) with zero compilation errors.
npm run build	Generates the optimized production client bundle in the dist/ directory.